DECK 09 · WEEK 5 · MORNING 1

Corporate Governance & WACC

Who owns the firm, who runs it, who finances it — and at what cost

CFA Level I · Vol 3 Learning Modules 1, 2, 3, 6

Roadmap — Corporate Governance, Capital Structure & Cost of Capital

5 sections · ~2 hours · Built bottom-up from CFA readings

01

Organizational forms & ownership

Sole proprietor · partnership · corporation · limited liability · Simon Property Group illustration

02

Investors & other stakeholders

Debt vs equity claims · ROE-with-leverage worked example · stakeholder map · ESG materiality

03

Corporate governance

Principal-agent · agency-cost worked example · dual-class · 3 core committees · Theranos / VW / Kobe vignettes

04

Capital structure

WACC mechanics · MM I and II · Gerhardt worked example · static trade-off · pecking order

05

Cost of capital wrap

CAPM for equity · Hamada lever/unlever beta · after-tax cost of debt · Aurelia full WACC

Learning objectives

By the end of this deck you should be able to:

1. Compare the three legal organizational forms of business on five attributes — legal identity, owner-manager relationship, owner liability, taxation, and access to financing.

2. Contrast the financial claims of debt and equity holders and quantify how financial leverage amplifies both shareholder return and risk.

3. Map a corporation's eight stakeholder groups, explain the principal-agent conflicts between them, and describe the governance mechanisms that mitigate each.

4. Compute a company's weighted-average cost of capital using market-value weights, an after-tax cost of debt, and a CAPM-based cost of equity.

5. Apply the Modigliani-Miller propositions (with and without taxes), the static trade-off theory, and the pecking order theory to real capital-structure decisions.

6. Lever and unlever equity beta using the Hamada formula to move between a comparable firm's observed risk and the target firm's own capital structure.

SECTION 01

Organizational forms & ownership

Three legal forms of business trade off simplicity for access to capital — sole proprietor, partnership, corporation

The corporate form is dominant globally by revenues and assets because it is the only form that offers limited liability, owner-manager separation, and frictionless ownership transfer simultaneously.

Attribute	Sole proprietor	General / Limited partnership	Corporation (public limited)
Legal identity	Extension of owner	Extension of partner(s)	Separate legal entity
Owner-manager	Owner operates	Partners operate (GP in LP)	Board & hired management
Owner liability	Unlimited	GP unlimited; LP limited	Limited to invested capital
Taxation	Pass-through (personal)	Pass-through (personal)	Corporate + dividend (double)
Access to financing	Owner + personal loans	Partners + partnership loans	Unbounded — public markets

WHY CORPORATIONS WIN ON SCALE

A share is the smallest possible ownership unit, transferable without disrupting operations, and carries a cap on loss equal to its purchase price. Together, these features let a firm pool capital from thousands of investors who demand no role in management.

Source: CFA Institute, Curriculum Vol 3, Learning Module 1, pp. 6–14.

Corporations combine limited liability with owner-manager separation — the combination that unlocks deep capital markets

Shareholders → elect → Board of Directors → hires → Management. Three layers, each acting (in theory) in the interests of those who elected them.

FIVE KEY FEATURES OF CORPORATIONS

1. Separate legal entity — can contract, sue, be sued, borrow
2. Limited shareholder liability — cap on loss = purchase price
3. Separation of ownership & management — elect board; board appoints CEO
4. Transferable shares — small, standardized ownership units
5. Perpetual life — survives any individual owner's exit

Drawback: double taxation (corporate + dividend).

SIMON PROPERTY GROUP — A HYBRID STRUCTURE

Simon owns USD 33B of retail real estate via two layers:

- Simon Property Group L.P. (limited partnership) holds the assets.
- Simon Property Group Inc. (corporation) is the General Partner with 87% of the LP.

The LP shields Inc. shareholders from real-estate liability while keeping Inc.'s own shareholders behind limited liability. A GP that is itself a corporation — a common structure for REITs and fund sponsors.

Source: CFA Institute, Curriculum Vol 3, Learning Module 1, Example 1 (Simon Property Group), pp. 15–20.

Public companies trade liquidity and price transparency for disclosure, listing costs, and the scrutiny of outside shareholders

The decision to go public (or to go private) is ultimately a trade between access to capital and freedom from continuous public scrutiny.

Dimension	Public (listed)	Private
Liquidity	Continuous exchange trading	Illiquid — negotiated block trades
Price transparency	Real-time quote, regulated disclosure	Negotiated, stale, often opaque
Disclosure burden	Quarterly 10-Q / interim audit, 10-K / annual	Bilateral with investors; often confidential
Access to capital	Broad — public debt & equity markets	Narrow — relationship-based
Typical shareholder base	Thousands to millions (retail + institutional)	Founders, PE, strategic investors, family
Governance scrutiny	Activists, proxy fights, say-on-pay votes	Insulated from public governance pressure

MOVING BETWEEN THE TWO

Private → Public: IPO · direct listing · SPAC reverse merger. Public → Private: LBO · MBO · take-private tender offer.

Source: CFA Institute, Curriculum Vol 3, Learning Module 1, §4 "Publicly vs. Privately Owned Corporate Issuers", pp. 21–30.

SECTION 02

Investors & other stakeholders

Debt is a fixed, priority, finite claim; equity is a residual, discretionary, permanent claim — the two are economically asymmetric

Debt investors are paid in full or not at all; equity investors get whatever is left. Same cash-flow pool, fundamentally different payoff profile.

Attribute	Debt	Equity
Tenor	Finite (e.g. 3m, 5y, 30y)	Indefinite / permanent
Claim type	Fixed, priority	Residual, discretionary
Payment	Contractual interest + principal	Dividend if board declares — or buyback
Return potential	Capped at promised rate	Theoretically unlimited upside
Maximum loss	Initial investment	Initial investment (limited liability)
Tax treatment of payout	Interest is tax-deductible for issuer	Dividends paid from after-tax profit
Voting rights	None	Yes — board elections, M&A, etc.
Issuer risk	Higher (obligation, bankruptcy)	Lower (no repayment obligation)

Source: CFA Institute, Curriculum Vol 3, Learning Module 2, pp. 39–47.

A worked ROE example shows how financial leverage amplifies both return and risk — the fundamental debt-equity trade-off

Same operating business, two capital structures, three revenue scenarios. Leverage turns a 30% return on assets into 60% ROE — and a 20% revenue drop into a −20% loss.

Scenario	Equity-financed (100% E)	Debt-financed (75% D, 25% E, interest 15)	ROE gap
Base: Rev 100, OpEx 70	NI 30 / E 100 = 30%	NI 30−15=15 / E 25 = 60%	+30pp
Upside: Rev 120 (+20%)	NI 50 / E 100 = 50%	NI 50−15=35 / E 25 = 140%	+90pp
Downside: Rev 80 (−20%)	NI 10 / E 100 = 10%	NI 10−15=−5 / E 25 = −20%	−30pp

INTERPRETATION

Leverage works when the firm's return on assets (30%) exceeds its cost of debt (15/75 = 20%). In the upside, leverage amplifies the 50% all-equity return to 140% on equity. In the downside, the same mechanism turns a positive 10% return into a net loss and destroys ~80% of book equity. A higher tax rate further favours debt (interest is tax-deductible), which is why capital-intensive firms lever up. Volatility, not level, is what debt truly amplifies.

Source: CFA Institute, Curriculum Vol 3, Learning Module 2, Example 1 paraphrased, pp. 42–44.

Eight stakeholder groups each bring a distinct set of claims, incentives, and conflict potential with the corporation

Shareholder-theory focuses only on residual claimants. Stakeholder-theory broadens the lens to every group with a vested interest — ESG analysis is a direct implication.

Stakeholder	What they provide	What they seek from the firm
Shareholders	Equity capital	Residual return, voting rights, capital appreciation
Debtholders (banks & bond-)	Debt capital	Timely interest & principal; covenant protection
Board of directors	Stewardship & oversight	Fiduciary duty to shareholders, fee, reputation
Managers	Human capital & strategy	Compensation, status, tenure, career options
Employees	Labour, skills, human capital	Fair pay, safety, development, job security
Customers	Revenue	Value, safety, service, stable supply
Suppliers	Inputs & trade credit	Timely payment; long-term relationship
Governments / society	Rule of law, infrastructure	Tax revenue, compliance, externality management

Source: CFA Institute, Curriculum Vol 3, Learning Module 2, Exhibit 6, pp. 48–54.

ESG factors are increasingly quantifiable and material — analysts bake them into cash-flow forecasts and discount rates

Vale's Brumadinho dam (2019), Equifax's 2017 breach, and Siemens' pre-2008 bribery culture all show how ESG mismanagement feeds directly into share-price drops and credit downgrades.

ENVIRONMENTAL

Climate (physical & transition risk) · pollution & waste · resource & land use · biodiversity · stranded assets

Material for utilities, O&G, mining, autos, airlines, food.

SOCIAL

Human capital · data privacy · product safety · community relations · labour standards

Material for tech, healthcare, consumer, pharma.

GOVERNANCE

Board independence · audit committee · executive pay · voting rights · bribery & corruption

Material for ALL issuers — and typically the most quantifiable of the three.

Vignette	ESG category	Financial impact
Vale — Brumadinho dam, 2019	Environmental	270 deaths · USD 7B settlement · debt briefly to speculative grade
Equifax — data breach, 2017	Social	140M records stolen · hundreds of USD M in costs · multi-year lawsuits
Siemens — bribery, pre-2008	Governance	EUR 3B+ fines & disgorgement · board + CEO replaced · policy overhaul

SECTION 03

Corporate governance

The corporation is a chain of principal-agent relationships — every link opens a potential conflict of interest and agency cost

Shareholders (principal) elect directors (agent); directors (now principal) hire managers (agent); each agent has information the principal does not. Agency costs fill the gap.

Agency cost = Direct monitoring cost + Forgone value (indirect)

1. **Insufficient effort**
   Managers shirk — too little monitoring, avoid hard decisions, distracted by side projects or charitable activities.
2. **Inappropriate risk appetite**
   All-cash pay → too risk-averse. Options-heavy pay → excessive risk-taking (asymmetric upside for the manager).
3. **Empire building**
   Compensation tied to firm size ⇒growth for growth's sake — acquisitions that destroy value but enlarge the CEO's fiefdom.
4. **Entrenchment**
   Managers protect their jobs — poison pills, staggered boards, copying peers to avoid standing out, overly complex transactions.
5. **Self-dealing**
   Corporate jets, club memberships, personal security, related-party transactions — cost borne by shareholders, enjoyed by the CEO.

Source: CFA Institute, Curriculum Vol 3, Learning Module 3, pp. 70–76.

Worked agency-cost example quantifies the cost of a corporate-jet habit — translating soft governance talk into a number

Perks that look small on a P&L capitalise into surprisingly large destroyers of shareholder value. A governance problem is a valuation problem.

ATLAS INDUSTRIAL — CEO PRIVATE-JET PERK

Atlas CEO uses a company-leased private jet for personal travel. The board does not require him to reimburse. Atlas' cost of equity is 10%; marginal tax rate 25%. Estimate the present value cost to shareholders — the agency cost of debt-free empire-building.

1. Annual jet cost: lease USD 3.2M + fuel & crew USD 0.8M = USD 4.0M pre-tax
2. After-tax cost to shareholders: 4.0 × (1 − 0.25) = USD 3.0M per year
3. Capitalize the perpetuity at cost of equity 10%: PV = 3.0 / 0.10 = USD 30M
4. If Atlas has 15M shares outstanding, the perk destroys USD 2.00 per share of intrinsic value
5. Board adds a clause requiring CEO to reimburse personal flights at charter rates → USD 2.5M/yr recovered → USD 25M of value restored

PV agency cost ≈ USD 30M — a governance fix with zero operational downside.

Source: Illustrative firm ("Atlas Industrial"); agency-cost framework from CFA Vol 3 LM 3, pp. 70–76.

Concentrated ownership, dual-class shares, and staggered boards each shift power toward insiders — sometimes at the expense of minority holders

CFA Institute has long advocated against dual-class structures — they hand one group disproportionate power over the will of the majority.

DISPERSED OWNERSHIP

No single shareholder controls. Main agency conflict is shareholder ↔ manager. Activists can force change via proxy contest.

CONCENTRATED OWNERSHIP

Founder / family / PE holds > 30%. Main conflict shifts to majority ↔ minority shareholders. Buffers the firm from hostile takeover.

DUAL-CLASS SHARES

Class A (public): 1 vote / share. Class B (insiders): 10+ votes / share. Small economic stake controls corporate destiny.

MAGNA INTERNATIONAL — HOW 3% CAN CONTROL 75%

Frank Stronach's family owned ~3% of Magna's total shares but 75% of voting rights via Class B shares (500 votes/share).

Shareholders bought the founder out for CAD 870M over 5 years to collapse the dual-class structure.

Source: CFA Institute, Curriculum Vol 3, Learning Module 3, "Controlling and Minority Shareholder Relationships", pp. 72–74.

Governance mechanisms divide into five categories — each addressing a specific principal-agent conflict

Reporting shrinks information asymmetry; shareholder & creditor mechanisms police the contract; board & government mechanisms set the rules of the game.

Category	Primary tools	Targets which conflict?
Reporting & transparency	Audited F/S · proxy · disclosures · IR	Information asymmetry
Shareholder mechanisms	AGM · EGM · proxy contest · say-on-pay · derivative lawsuit · activism	Manager ↔ shareholder
Creditor mechanisms	Bond indenture · covenants · ad hoc committee · trustee	Shareholder ↔ debtholder
Board & management	Audit / Nom / Comp committees · LTIPs · independent chairman	Manager ↔ shareholder
Government / regulator	Securities law · comply-or-explain codes · listing rules · SOX	Issuer ↔ society

COMPLY-OR-EXPLAIN

Most jurisdictions (UK, EU, Japan) use a "comply-or-explain" model — issuers either adopt the code or publicly justify departures. The US relies instead on SEC securities law, Delaware corporate law, and exchange listing rules.

Source: CFA Institute, Curriculum Vol 3, Learning Module 3, "Corporate Governance Mechanisms", pp. 76–84.

Three core board committees — Audit, Nominating/Governance, Compensation — embed independence where it matters most

Each committee reports to the full board but does the first-level work. Best practice on all three: majority independent directors; on Audit and Compensation, 100%.

AUDIT COMMITTEE

Mandate: financial reporting integrity, internal controls, external auditor selection & fees, whistleblower channel.

Best practice:
- 100% independent directors
- ≥ 1 with accounting / financial expertise
- Private sessions with auditor (no management)

Red flag: CFO or insider on committee.

NOMINATING / GOVERNANCE

Mandate: director selection, board composition, succession planning, corporate governance code, conflicts-of-interest policy.

Best practice:
- Majority independent
- Explicit skills matrix
- Annual board evaluation

Red flag: CEO handpicking directors.

COMPENSATION

Mandate: executive pay plan, performance metrics, equity grants, say-on-pay voting.

Best practice:
- 100% independent directors
- LTIP with multi-year vesting
- Clawback provision
- Peer-group benchmarking

Red flag: CEO-only metrics, single-year cash bonus.

Source: CFA Institute, Curriculum Vol 3, Learning Module 3, Exhibit 2, pp. 80–82.

Strong governance lowers the cost of capital — the Theranos, Volkswagen, and Kobe Steel cases show the price of getting it wrong

Every failure traces back to one of three root causes: inadequate oversight, captive board, or misaligned compensation.

THERANOS (USA, 2015-18)

Governance failure — celebrity board with no medical expertise dismissed whistleblowers.

- Peak valuation: USD 10B (2014)
- Final valuation: USD 0 (2018)
- CEO convicted of fraud (2022)

Root cause: inadequate oversight.

VOLKSWAGEN (DE, DIESELGATE)

Governance failure — illegal emission "defeat devices" installed in 11M vehicles.

- Stock fell > 33% in days
- EUR 32B in fines, recalls, legal costs
- CEO indicted; criminal charges in US + DE

Root cause: captive supervisory board dominated by founders.

KOBE STEEL (JP, 2017-18)

Governance failure — decades of falsified quality data for steel, copper, aluminium products.

- Stock to 5-year low
- Cost of debt to record highs
- Board restructured: independent chair, 3 independent directors on audit committee

Root cause: profit-over-quality culture, no quality-control oversight.

[1] Academic evidence: Gompers, Ishii & Metrick (2003) find a governance index explains a meaningful share of firm valuations; Bhojraj & Sengupta (2003) link governance to credit ratings and bond yields.

Source: CFA Institute, Corporate Finance, Q3 3, Learning Objectives, case vignettes, pp. 85–91.

SECTION 04

Capital structure

WACC blends the after-tax cost of debt and the cost of equity — the single most important discount rate in corporate finance

WACC is simultaneously (i) the firm's discount rate for NPV analysis, (ii) the hurdle rate for IRR, and (iii) the investors' required return weighted by capital structure.

$$WACC = w_d \cdot r_d \cdot (1 - t) + w_e \cdot r_e$$

where: w_d, w_e = market-value (or target) weights of debt & equity; r_d = pre-tax cost of debt (YTM); r_e = cost of equity (CAPM); t = marginal tax rate.

QUICK CHECK — ABC CORPORATION

ABC: 40% debt / 60% equity at market value. Pre-tax r_d = 4%. r_e = 10%. Marginal tax rate t = 23%.

1. After-tax cost of debt: 4% × (1 − 0.23) = 3.08%
2. Debt contribution: 0.40 × 3.08% = 1.232%
3. Equity contribution: 0.60 × 10% = 6.00%
4. WACC = 1.232% + 6.00% = 7.23%

WACC = 7.23% — the rate at which ABC must discount projects of average firm risk.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, "The Cost of Capital", pp. 178–182.

Full WACC worked example — converting book weights to market weights, computing after-tax debt, and assembling the WACC

Market values reflect investors' current opportunity cost; book values reflect historic accounting. For cost-of-capital work, market always wins.

Meridian Corp — capital stack	Book value	Market value	BV weight	MV weight
Long-term debt (5.0% coupon, 7y, YTM 4.0%)	400	420	33%	26%
Common equity (60M shares × USD 20)	800	1,200	67%	74%
Total capital	**1,200**	**1,620**	**100%**	**100%**

ASSEMBLE THE WACC — MARGINAL TAX RATE 25%

Pre-tax cost of debt = YTM = 4.0%. Cost of equity (CAPM) = 9.5%. Use market-value weights.

1. After-tax cost of debt: 4.0% × (1 − 0.25) = 3.00%
2. Debt contribution: 0.26 × 3.00% = 0.78%
3. Equity contribution: 0.74 × 9.50% = 7.03%
4. WACC = 0.78% + 7.03% = 7.81% at market weights (vs 7.33% at book weights — book understates WACC here)

WACC ≈ 7.8% — use for projects of average Meridian risk; adjust up for higher-risk projects.

Source: Illustrative firm ("Meridian Corp"); framework from CFA Vol 3 LM 6, pp. 178–182.

Capital structure depends on business model and corporate life cycle — startup, growth, and mature firms finance themselves differently

Business risk falls and free cash flow rises as firms mature — lenders respond by offering bigger cheques at lower spreads, on less restrictive terms.

Life-cycle feature	Startup	Growth	Mature
Revenue growth	Initial / zero	Rising fast	Slowing, predictable
Free cash flow	Negative	Negative / turning positive	Stable positive
Business risk	High	Medium	Low
Debt availability	Little to none	Rising	High
Typical debt type	Convertible · leases	Secured term loans	Unsecured bonds, IG-rated
Typical equity source	Founders · VC	Growth equity · late-stage PE	Public market · retained earnings

CAPITAL-LIGHT VS CAPITAL-INTENSIVE CUTS ACROSS ALL STAGES

Uber / Airbnb are capital-light networks; utilities, semis, airlines are capital-intensive. The intensity axis drives HOW MUCH capital; the life-cycle axis drives WHAT TYPE.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, Exhibit 2, pp. 183–186.

MM Proposition I: in a frictionless world, capital structure is irrelevant — firm value is the PV of cash flows, full stop

Modigliani & Miller (1958) — the starting point of modern corporate finance, and the benchmark against which every real-world deviation is measured.

$$V_L = V_U \quad \text{— firm value independent of D/E mix}$$

where: V_L = value of levered firm; V_U = value of unlevered firm. Holds under the five MM assumptions below.

MM Assumption	What it rules out in the real world
1. Homogeneous expectations	Investors disagree on cash-flow forecasts in reality.
2. Perfect capital markets (no tax, no friction)	Interest is tax-deductible, bankruptcy has direct & indirect costs.
3. Common risk-free borrow/lend rate	Individuals borrow at higher rates than corporations.
4. No agency costs	Managers don't always act in shareholder interest (see Deck §3).

HOMEMADE LEVERAGE

An investor preferring 70% leverage at a 50%-levered firm can borrow on personal account to replicate it. Arbitrage forces $V_L = V_U$.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, "MM Proposition I without Taxes", pp. 193–195.

MM Proposition II without taxes: more debt raises the cost of equity 1-for-1, leaving WACC unchanged

The cheaper debt is exactly offset by a rising cost of equity — the classic MM see-saw.

$$r_e = r_0 + (r_0 - r_d) \cdot D / E$$

where: r_e = cost of levered equity; r_0 = cost of unlevered equity (= WACC when D = 0); r_d = cost of debt; D/E = debt-to-equity ratio.

GERHARDT CORP — ALL-EQUITY → BUYBACK WITH DEBT

Gerhardt: expected perpetual CF = EUR 5,000; cost of equity (= WACC) = 10%; no taxes. Firm value = 5,000 / 0.10 = EUR 50,000. Issue EUR 15,000 debt at 5% to repurchase equity.

1. Post-buyback: D = 15,000; E = 50,000 − 15,000 = 35,000; D/E = 0.4286
2. New cost of equity: r_e = 10% + (10% − 5%) × (15,000 / 35,000) = 12.143%
3. New WACC: (15/50)(5%) + (35/50)(12.143%) = 1.50% + 8.50% = 10.00%
4. Firm value: still 5,000 / 0.10 = EUR 50,000 — unchanged. ✓

WACC = 10% before and after. Cheaper debt offset exactly by a higher cost of equity.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, Example 2 (Gerhardt), pp. 195–197.

MM with taxes: the interest tax shield makes debt value-creating — $V_L = V_U + tD$, WACC falls as leverage rises

Once we let governments tax corporate profits, every dollar of deductible interest is a dollar the firm keeps. Capital structure starts to matter.

$$V_L = V_U + t \cdot D \qquad r_e = r_0 + (r_0 - r_d)(1 - t) \cdot D / E$$

where: t = marginal corporate tax rate; t·D = present value of the debt tax shield (assumes perpetual debt).

GERHARDT CORP REDUX — TAX RATE 25%

Same cash flow (EUR 5,000) and cost of capital assumptions, now with corporate tax at 25%. $V_U = (5{,}000 \times 0.75) / 0.10 =$ EUR 37,500. Issue EUR 15,000 debt at 5%.

1. PV of tax shield: $t \cdot D = 0.25 \times 15{,}000 =$ EUR 3,750
2. $V_L = V_U + t \cdot D = 37{,}500 + 3{,}750 =$ EUR 41,250; $E = 41{,}250 - 15{,}000 = 26{,}250$
3. $r_e = 10\% + (10\% - 5\%)(1 - 0.25)(15{,}000 / 26{,}250) = 12.143\%$
4. $WACC = (15/41.25)(5\%)(0.75) + (26.25/41.25)(12.143\%) = 1.36\% + 7.73\% = 9.091\%$

WACC falls from 10% → 9.09%; firm value rises EUR 3,750. Debt creates value equal to the tax shield.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, Example 3 (Gerhardt with taxes), pp. 197–200.

Static trade-off theory: the optimal capital structure balances the tax shield against the present value of financial-distress costs

Pure MM with taxes implies 100% debt is optimal. Bankruptcy costs cap that: beyond some D, expected distress costs outweigh further tax shield.*

$$V_L = V_U + t \cdot D - PV(\text{financial-distress costs})$$

Debt D (USD M)	Tax shield t·D (M)	PV distress costs (M)	V_L (M)	Δ vs V_U
0	0.0	0.0	45.0	—
5	1.5	0.1	46.4	+1.4
10	3.0	0.2	47.8	+2.8
15	4.5	0.5	49.0	+4.0
20 ← D*	6.0	1.5	49.5	+4.5
25	7.5	6.0	46.5	+1.5
30	9.0	12.0	42.0	−3.0

Boulder Inc. (V_U = 45, t = 30%): D* = USD 20M maximises V_L at USD 49.5M. Beyond this point, distress costs dominate.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, Practice Problem (Boulder Inc.), pp. 203–212.

Pecking order theory: information asymmetry pushes managers to prefer retained earnings → debt → equity, in that order

Equity issuance signals to the market that management thinks shares are overpriced — so managers avoid it unless they must. Static trade-off sets a target; pecking order explains deviations.

1. INTERNAL FUNDS

Retained earnings. No floatation cost, no signal, no disclosure.

First choice because invisible to outsiders.

2. DEBT

If internal funds insufficient, borrow.

Debt issuance is mildly POSITIVE signal — commits firm to future payments, so management must be confident.

3. EQUITY

Last resort. Equity issuance is NEGATIVE signal — existing owners only share an overpriced asset.

Empirically: equity-issuing firms underperform for 3–5 years.

JENSEN (1986) — FREE-CASH-FLOW HYPOTHESIS

Debt disciplines managers. Required interest and principal payments force efficient use of cash and deter empire-building acquisitions. Leveraged buyouts of cash-rich mature firms (think 1980s RJR Nabisco) exploit this mechanism directly — replace discretionary cash with contractual debt service.

Source: Myers & Majluf (1984); CFA Institute, Curriculum Vol 3, Learning Module 6, pp. 205–208.

SECTION 05

Cost of capital wrap

CAPM gives the cost of equity as the risk-free rate plus beta times the equity risk premium — the industry-standard equity discount rate

CAPM: the only systematic (market) risk is compensated; idiosyncratic risk is diversified away. Beta = sensitivity to market returns.

$$r_e = r_f + \beta \cdot (r_m - r_f) = r_f + \beta \cdot ERP$$

where: r_f = risk-free rate (typically 10y Treasury); β = levered equity beta; ERP = equity risk premium (historical 4–6% in developed markets).

COST-OF-EQUITY FOR A TYPICAL US LARGE-CAP

Assumptions: r_f = 4.0% (10y UST); ERP = 6.0% (long-run US historical); levered β = 1.20 (e.g. an industrial manufacturer).

1. Market risk premium (MRP) term: β × ERP = 1.20 × 6.0% = 7.20%
2. Cost of equity: $r_e = r_f + \beta \cdot ERP$ = 4.0% + 7.20%
3. r_e = 11.20%
4. Sensitivity: at β = 1.50, r_e = 13.00%; at β = 0.80, r_e = 8.80%.

r_e = 11.2% — use directly in WACC if the firm's actual leverage equals the peer group's.

Source: CFA Institute, Curriculum Vol 3, LM 6 (cost-of-equity discussion); CAPM developed in Vol 6 (Equity).

Lever and unlever beta to move between a comparable firm's observed risk and your target firm's own capital structure

Comparable-company beta reflects that company's leverage. Unlever (β_asset) to isolate business risk; then relever at the target firm's D/E.

$$\beta_U = \beta_L / [\,1 + (1 - t) \cdot D/E\,] \qquad \beta_L = \beta_U \cdot [\,1 + (1 - t) \cdot D/E\,]$$

where: β_U = unlevered (asset) beta; β_L = levered (equity) beta. Debt assumed to have beta ≈ 0 (investment-grade).

UNLEVER PEER, RELEVER TARGET

Peer firm's levered β = 1.20, D/E = 0.50, t = 25%. Target firm has D/E = 1.00 (more levered). Compute target's levered β.

1. Unlever peer: β_U = 1.20 / [1 + (1 − 0.25) × 0.50] = 1.20 / 1.375 = 0.873
2. Relever at target's D/E: β_L,target = 0.873 × [1 + (1 − 0.25) × 1.00] = 0.873 × 1.75
3. β_L,target = 1.527
4. At r_f = 4%, ERP = 6%: target r_e = 4% + 1.527 × 6% = 13.16%

Target β rises from 1.20 (peer) to 1.53 (target) because target is more levered. Cost of equity rises 1.96 pp.

Source: Hamada (1972); CFA Vol 3 LM 6 context on levered vs unlevered returns, pp. 195–200.

The after-tax cost of debt is the yield-to-maturity scaled by (1 − t); it is the YTM, not the coupon, that matters

The coupon is historic — it locks in the cash interest. The YTM is what the firm would pay to raise new debt today, which is what discounting requires.

$$\text{After-tax } r_d = \text{YTM} \cdot (1 - t)$$

where: Use the YTM on the firm's own currently traded bonds; if illiquid, match by rating, sector, and maturity.

Credit / tenor (indicative)	Pre-tax YTM	After-tax @ t = 25%	After-tax @ t = 35%
5y, investment-grade A (mature industrial)	4.2%	3.15%	2.73%
7y, BBB (capital-intensive industrial)	5.4%	4.05%	3.51%
10y, BB (leveraged buyout)	6.8%	5.10%	4.42%
10y, B (speculative-grade)	8.5%	6.38%	5.53%
30y, CCC (distressed)	11.0%	8.25%	7.15%

WHY YTM, NOT COUPON

A bond issued at 5% coupon five years ago may trade at YTM 7% today. The 7% reflects what a new marginal investor demands — and therefore what the firm would pay on new debt.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, pp. 178–180.

Worked WACC wrap-up — Aurelia Energy Corp pulls CAPM, Hamada, and after-tax debt into a single discount rate

Five sequential steps: identify peer, unlever peer beta, relever at target D/E, run CAPM, combine with after-tax debt.

AURELIA ENERGY CORP — FULL WACC

Mid-cap independent power producer. Target capital structure: 40% debt / 60% equity (D/V = 0.4, D/E = 0.667). Marginal tax rate 25%. r_f = 4.0%, ERP = 6.0%. Peer utility trades at β_L = 0.95, D/E = 0.80. Aurelia's 10y unsecured bond YTM = 5.2%.

1. 1. Unlever peer beta: $\beta_U = 0.95 / [1 + (1 - 0.25)(0.80)] = 0.95 / 1.60 = 0.594$
2. 2. Relever at Aurelia's D/E = 0.667: $\beta_L = 0.594 \times [1 + (1 - 0.25)(0.667)] = 0.594 \times 1.50 = 0.891$
3. 3. CAPM cost of equity: $r_e = 4.0\% + 0.891 \times 6.0\% = 4.0\% + 5.35\% = 9.35\%$
4. 4. After-tax cost of debt: $r_d \cdot (1 - t) = 5.2\% \times 0.75 = 3.90\%$
5. 5. WACC = $w_d \cdot r_d (1-t) + w_e \cdot r_e = 0.40 \times 3.90\% + 0.60 \times 9.35\% = 1.56\% + 5.61\% = 7.17\%$

Aurelia WACC ≈ 7.2% — apply for NPV of projects of average Aurelia risk; add a project-risk premium for riskier assets.

[1] Rule of thumb: CAPM + Hamada is the textbook default for public companies with comparable peers; for private companies, build-up method with size premium is more common.

Source: illustrative firm ("Aurelia Energy Corp"); synthesises frameworks from CFA Vol 3 LM 6.

KEY TAKEAWAYS

Deck 9 — Corporate Governance, Capital Structure & Cost of Capital

- Three legal forms of business differ on 5 attributes — liability, tax, access to finance, owner-manager relationship, legal identity. Corporations dominate because they combine limited liability, owner-manager separation, and frictionless ownership transfer.
- Every principal-agent link (shareholder→board, board→manager) creates potential agency costs: insufficient effort, risk mis-appetite, empire building, entrenchment, self-dealing. Governance mechanisms address each.
- WACC = $w_d \cdot r_d \cdot (1 - t) + w_e \cdot r_e$. Use MARKET-value weights; after-tax r_d uses YTM, not coupon; r_e from CAPM. Single most important rate in corporate finance.
- Static trade-off: optimal D* balances tax shield against PV of distress costs (inverted-U in V_L). Boulder: D* = 20 maximises V_L = 49.5. Pecking order: retained earnings → debt → equity, driven by information asymmetry.
- Debt and equity are economically asymmetric: debt is fixed, priority, finite, tax-deductible; equity is residual, discretionary, permanent. Leverage amplifies both ROE and ROE volatility — the same mechanism drives both.
- Three core committees (Audit, Nominating, Compensation) embed independence. Real-world failures — Theranos, VW Dieselgate, Kobe Steel — trace to inadequate oversight, captive boards, or misaligned pay.
- MM I & II (no tax): capital structure irrelevant — cheaper debt exactly offset by higher cost of equity. With taxes, $V_L = V_U + tD$; debt creates value via the tax shield.
- CAPM ($r_e = r_f + \beta \cdot ERP$) + Hamada ($\beta_U \leftrightarrow \beta_L$ via D/E and tax) give the full toolkit. Aurelia Energy: unlever peer β, relever at target D/E, combine with after-tax debt → WACC ≈ 7.2%.